

بسم الله الرحمن الرحيم

Proton $\rightarrow +e$
electron $\rightarrow -e$
neutron $\rightarrow 0$

electric charge

Groups

- ① 2 electrons and 6 protons: $4 \rightarrow 4 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-19} \text{ C}$
- ② 6 electrons and 2 protons: $4 \rightarrow -4 \times 1.6 \times 10^{-19} = -6.4 \times 10^{-19} \text{ C}$
- ③ 12 Protons and 2 neutrons $\rightarrow 12 \times 1.6 \times 10^{-19} = 19.2 \times 10^{-19} \text{ C}$
- ④ 12 electrons and 2 neutrons $\rightarrow 12 \times 1.6 \times 10^{-19} = -19.2 \times 10^{-19} \text{ C}$

سهمه الکترون

electric charge

Proton $\rightarrow +e$
electron $\rightarrow -e$
neutron $\rightarrow 0$

⑤ 6 Protons and 2 electrons $\rightarrow m = 6 \times 1.67 \times 10^{-27} + 2 \times 9.11 \times 10^{-31} = ?$

⑥ 6 Protons and 2 neutrons $\rightarrow m = 6 \times 1.67 \times 10^{-27} + 2 \times 1.67 \times 10^{-27} = ?$

Contact and separate

two charges: $q_1 = 2 \mu C$ and $q_2 = -8 \mu C$, Contact and separate charge of each.

$$\frac{-6}{2} = -3 \mu C$$

Three charges: $q_1 = 6 C$, $q_2 = -4 C$ and $q_3 = 10 C$

$$\frac{12}{3} = 4 C$$

Electric field

$q_1 = 10 \mu\text{C}$ and $q_2 = 5 \mu\text{C}$ separated by 30 cm in air

$$F = k_e \frac{q_1 q_2}{r^2} = 9 \times 10^9 \times \frac{10 \times 10^{-6} \times 5 \times 10^{-6}}{0.3^2} = \underline{5 \text{ N}}$$

$q_1 = 10 \mu\text{C}$ and $q_2 = -5 \mu\text{C}$ separated by 50 cm

$$F = 9 \times 10^9 \times \frac{10 \times 10^{-6} \times 5 \times 10^{-6}}{0.5^2} = 0.018 \text{ N}$$

$q_1 = 10 \mu\text{C}$ and $q_2 = -5 \mu\text{C}$ separated by 5 cm

$$F = 9 \times 10^9 \times \frac{10 \times 10^{-6} \times 5 \times 10^{-6}}{0.05^2} = \underline{72 \text{ N}}$$

$$k_e = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9$$

$$k_e = \frac{1}{4\pi\epsilon_0}$$

$$k_e = \frac{9 \times 10^9}{\epsilon_r}$$

↓
في الهواء

مساحة المجال الكهربائي

Electric field

A region around a charge.

⊕ $\vec{F} \Rightarrow \vec{E}$ Positive charge: $\vec{F}$ and $\vec{E}$

Same direction

$\vec{F} \Leftarrow \ominus$ Negative charge: $\vec{F}$ and $\vec{E}$

opposite direction

$$\frac{F}{q} = \frac{V}{l}$$

$$V_f^2 - V_i^2 = 2a \times l$$

$$V_f = \sqrt{2al} = \sqrt{2 \times 9 \times 0.02}$$

$$V_f = V_i + at$$

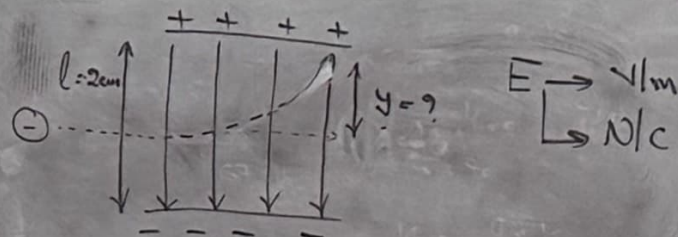
$$t = \frac{V_f}{a} = \frac{0.6}{9}$$

An electron ^{starts from rest} enters a box of width $l = 2\text{cm}$ has $E = 9000\text{N}$ as shown.
Find the displacement made by that electron.

$$a = \frac{qE}{m} = \frac{1.6 \times 10^{-19} \times 9000}{9.1 \times 10^{-31}}$$

$$\therefore y = V_i t + \frac{1}{2} at^2$$

$$y = \frac{1}{2} at^2$$



سواء ايجابية او سلبية

Electric field

A region around a charge.

⊕ $\vec{F} \Rightarrow \vec{E}$ Positive charge: $\vec{F}$ and $\vec{E}$

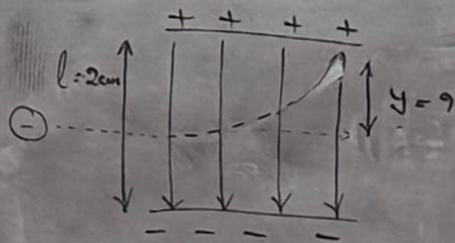
Same direction

$\vec{F} \Leftarrow \ominus \Rightarrow \vec{E}$ Negative charge: $\vec{F}$ and $\vec{E}$

opposite direction

$$\vec{E} = \frac{\vec{F}}{q} \rightarrow \vec{F} = q \vec{E}$$

Speed of light $c = \frac{E}{B}$



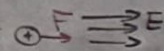
+ $-2C$ moves in $E = 30 \text{ N/C}$ at $\theta = 40^\circ$ with $+x$
Force on the electron is,

$$\vec{F} = q \vec{E} = 2 \times 30 = 60 \text{ N at } 90^\circ$$

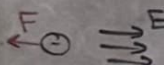
بسم الله الرحمن الرحيم

Electric field

A region around a charge.



Positive charge: F and E

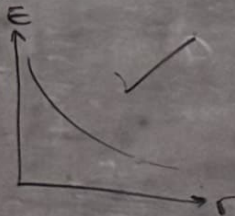


Negative charge: F and E

$$E = k \frac{Q}{r^2}$$

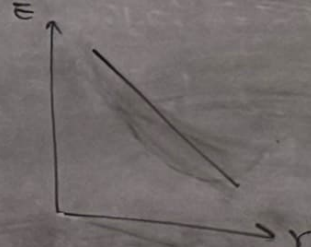
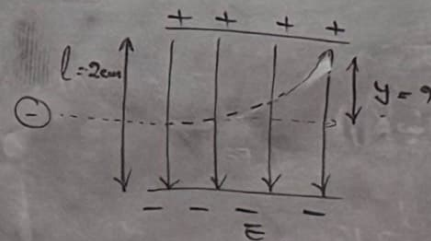
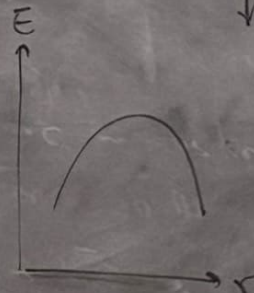
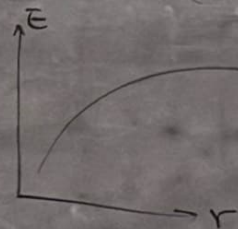
graphs

relation
between E and r



Same direction

Opposite direction



Electric field

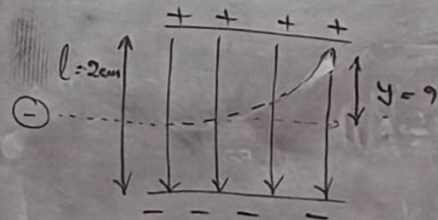
A region around a charge.

Positive charge: F and E

Same direction

Negative charge: F and E

Opposite direction



* $V_p = 2x^2 - 3x^3$

Find the electric field at $x = 2m$

$$E = 4x - 9x^2$$

$$= 4(2) - 9(2)^2 = 8 - 36 = -32 \text{ V/m}$$